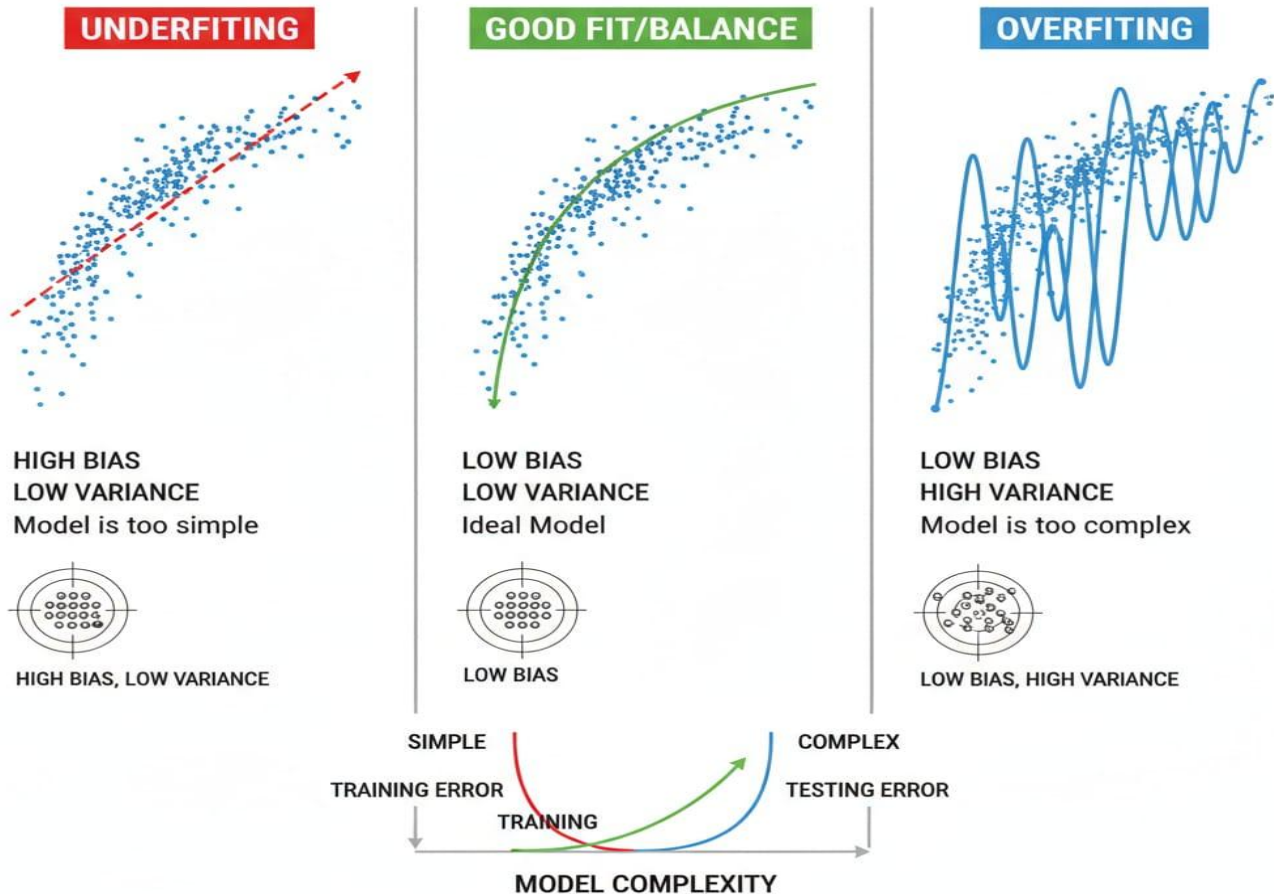


Fundamentals of AI and ML

Assignment

Q) Variance and Bias (Diagram, Overfit, Underfit)

BIAS-VARIANCE TRADE-OFF



- For the best-fit machine learning model, you should aim for low bias and low variance.
- This combination ensures the model accurately captures the underlying patterns (low bias) while remaining robust and generalizing well to unseen, new data (low variance).
- It acts as the sweet spot balancing underfitting and overfitting.

Bias

Bias is the error that occurs when a model is too simple to capture the true patterns in the data.

1. **High bias:** The model oversimplifies, misses patterns and underfits the data.
2. **Low bias:** The model captures patterns well and is closer to the true values.

Example: A neural network with too few layers or neurons fails to capture complex patterns, producing consistently inaccurate outputs. This is called underfitting.

Mathematically, the formula for bias is:

$$\text{Bias}^2 = (\mathbb{E}[\hat{f}(x)] - f(x))^2$$

Where,

- $\hat{f}(x)$: predicted value by the model
- $f(x)$: true value
- $\mathbb{E}[\hat{f}(x)]$: expected prediction over different training sets

Variance

Variance arises when a model becomes too sensitive to training data and it captures noises in data too. It fails to give prediction on unseen new data.

1. **High variance:** The model is too sensitive to small changes and may overfit.
2. **Low variance:** The model is more stable but might miss some patterns.

Example: A deep decision tree that memorizes the training data perfectly but performs poorly on new data shows high variance, this is known as overfitting.

Mathematically, the formula for variance is:

$$\text{Variance} = \mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]$$

Where,

- $\hat{f}(x)$: predicted value by the model
- $\mathbb{E}[\hat{f}(x)]$: average prediction over multiple training sets

Detailed Explanation

Why "Low Bias, Low Variance"?

- **Low Bias:** Implies the model does not make strong, incorrect assumptions about the data, allowing it to fit the training set closely.
- **Low Variance:** Implies the model is not overly sensitive to small fluctuations or noise in the training data, meaning it performs consistently on test data.
- **Best Fit:** A model with both low bias and low variance minimizes total error, avoiding the high error of underfitting (high bias) and the poor generalization of overfitting (high variance).

Understanding the Trade-off (Diagram Concept)

- High Bias (Underfit): The model is too simple (e.g., linear model on non-linear data), leading to high training and test errors.
- High Variance (Overfit): The model is too complex (e.g., high-degree polynomial), capturing noise as if it were a pattern. It performs great on training data but poorly on test data.
- Low Bias/Low Variance (Best Fit): The model is balanced, capturing the underlying pattern without modeling the noise.

Summary Table

Model State	Bias	Variance	Complexity
Underfit	High	Low	Low
Overfit	Low	High	High
Best Fit	Low	Low	Optimal